

**EEE 335 Laboratory 3
Report Submission**

Notes (same as Lab 2):

- Do not forget to copyright a curve-based image before you paste it in this report
- Only images that have curves need to be copyrighted.
- Circuit images do not need to be copyrighted if you plan to paste them in this report as long as the title bar has your name as you did in Lab 1
- Scanned documents of your handwritten analysis do not need to be copyrighted
- In this report, only copyright one curve-based image, not more. Choose ANY image in ANY task of your choice to copyright as long as it is an image containing some curve.
- Hand calculations asked in this report can be
 - written on a piece of paper, scanned and pasted, OR
 - typed on a computer, OR
 - provided in ANY other legible method

Task 1. Inverter Design and Simulation (15)

$$\mu_n \left(\frac{W_n}{L_n} \right) = \mu_p \left(\frac{W_p}{L_p} \right)$$

$$W_p = \frac{\mu_n}{\mu_p} W_n = \frac{274.75 \mu}{95.73 \mu} (450 n) = 1,291.25 nm$$

$$= 1.291 \mu m$$

$$t_{PHL} = \frac{\alpha_n C}{k_n \left(\frac{W}{L} \right) V_{DD}} \quad \alpha_n = \frac{1}{4} - \frac{3V_{thn}}{V_{DD}} + \left(\frac{V_{thn}}{V_{DD}} \right)^2$$

$$= \frac{1}{4} - \frac{3 \cdot 0.49}{2.5} + \left(\frac{0.49}{2.5} \right)^2$$

$$= 0.991$$

$$t_{PHL} = \frac{0.991 (1p)}{274.75 \mu (0.3 \mu / 0.3 \mu) 2.5}$$

$$= 1.443 ns$$

$$t_{PLH} = \frac{\alpha_p C}{k_p \left(\frac{W}{L} \right) V_{DD}} \quad \alpha_p = \frac{1}{4} - \frac{3|V_{thp}|}{V_{DD}} + \left(\frac{|V_{thp}|}{V_{DD}} \right)^2$$

$$= \frac{1}{4} - \frac{3 \cdot 0.49}{2.5} + \left(\frac{0.49}{2.5} \right)^2$$

$$= 1.087$$

$$= \frac{1.087 (1p)}{95.73 \mu (1.291 \mu / 0.3 \mu) 2.5}$$

$$= 1.754 ns$$

Scan 1. Inverter DC Hand Calculations

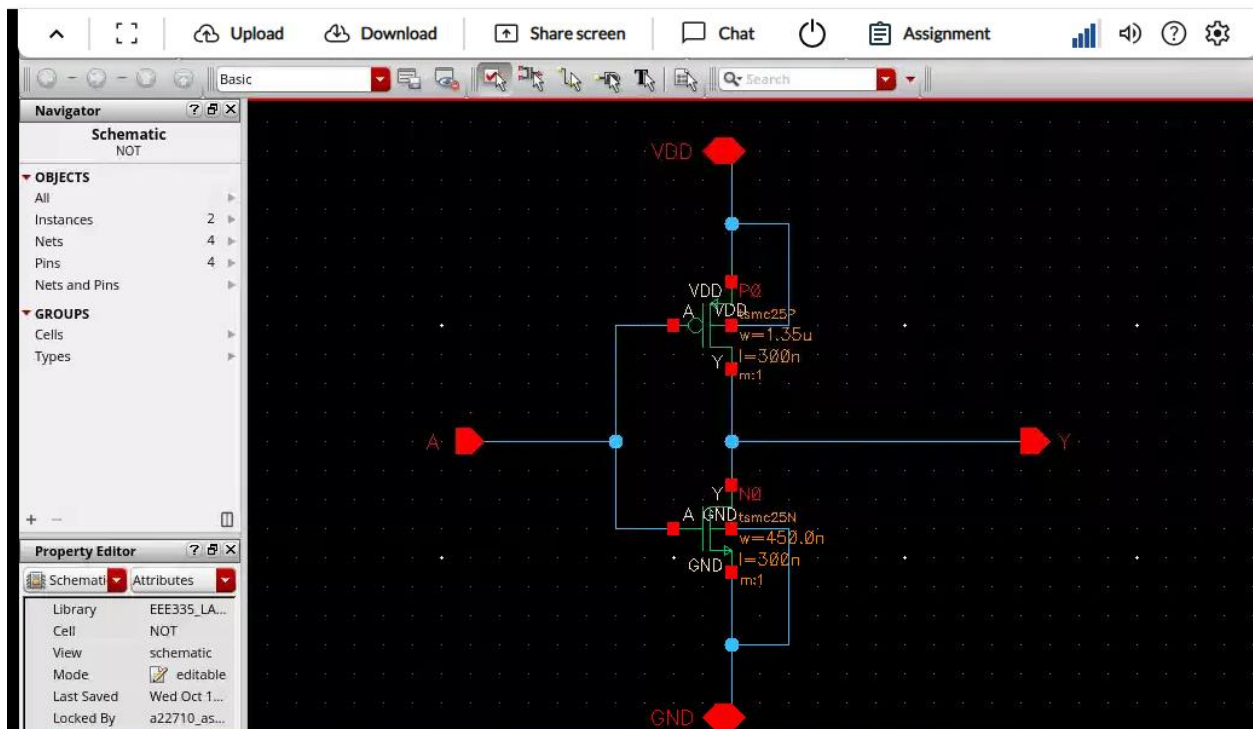


Fig. 1. Inverter (NOT) Schematic Cell

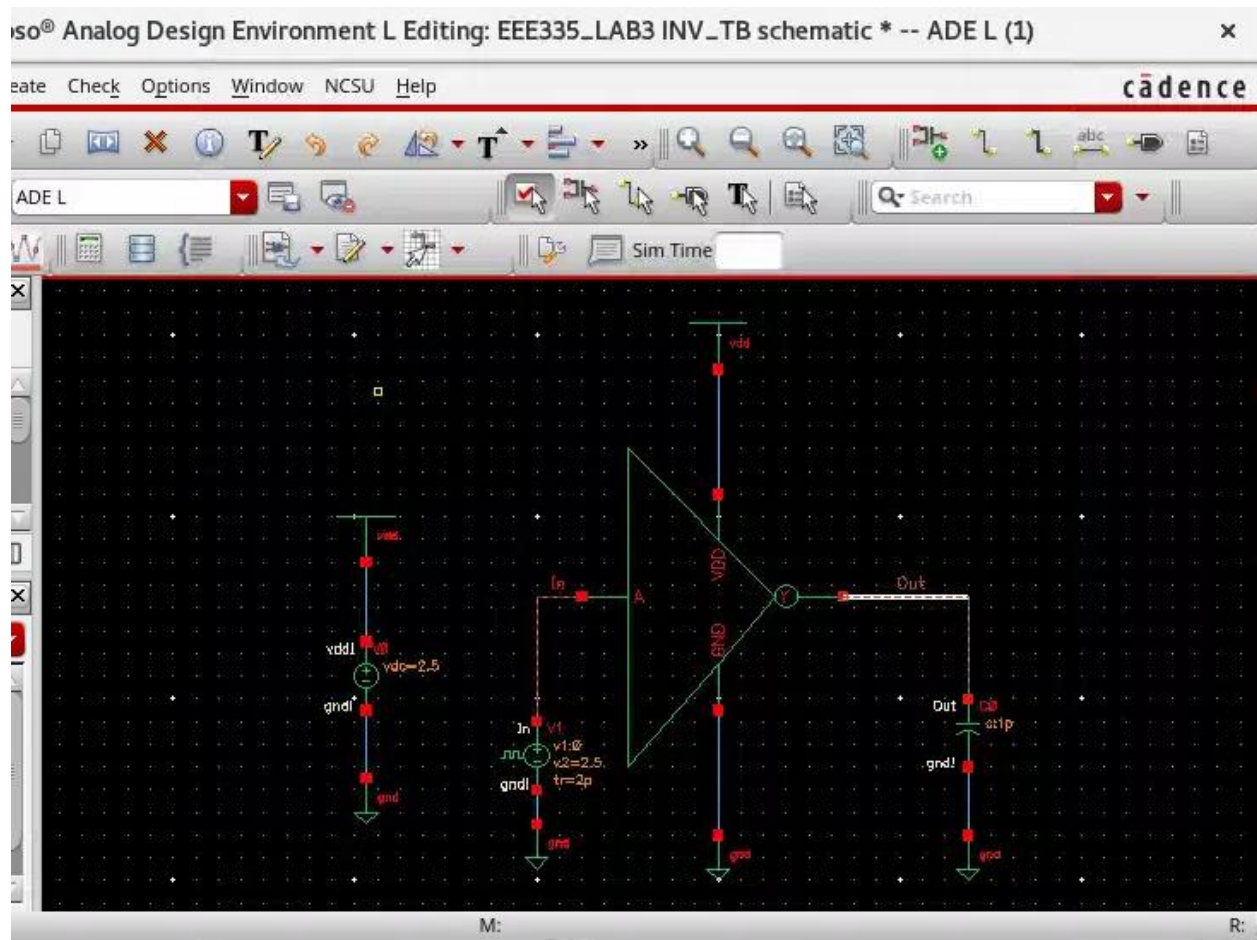


Fig. 2. NOT (Inverter) Test Bench Schematic

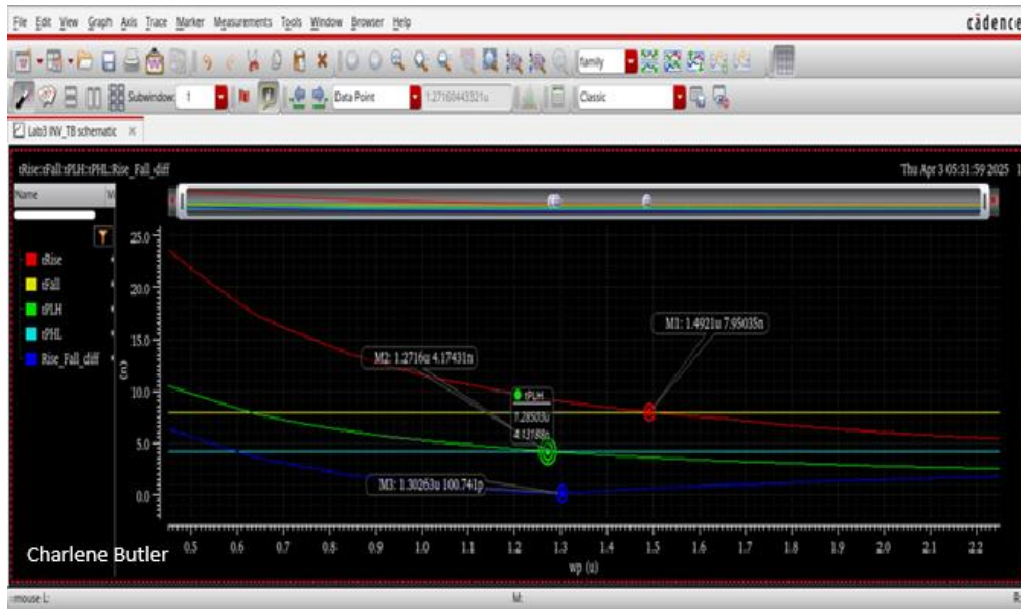


Fig. 3. Inverter Parametric Sweep of PMOS Width (t_{pHL} , t_{pLH} , t_r , t_f , Rise Fall Delay)

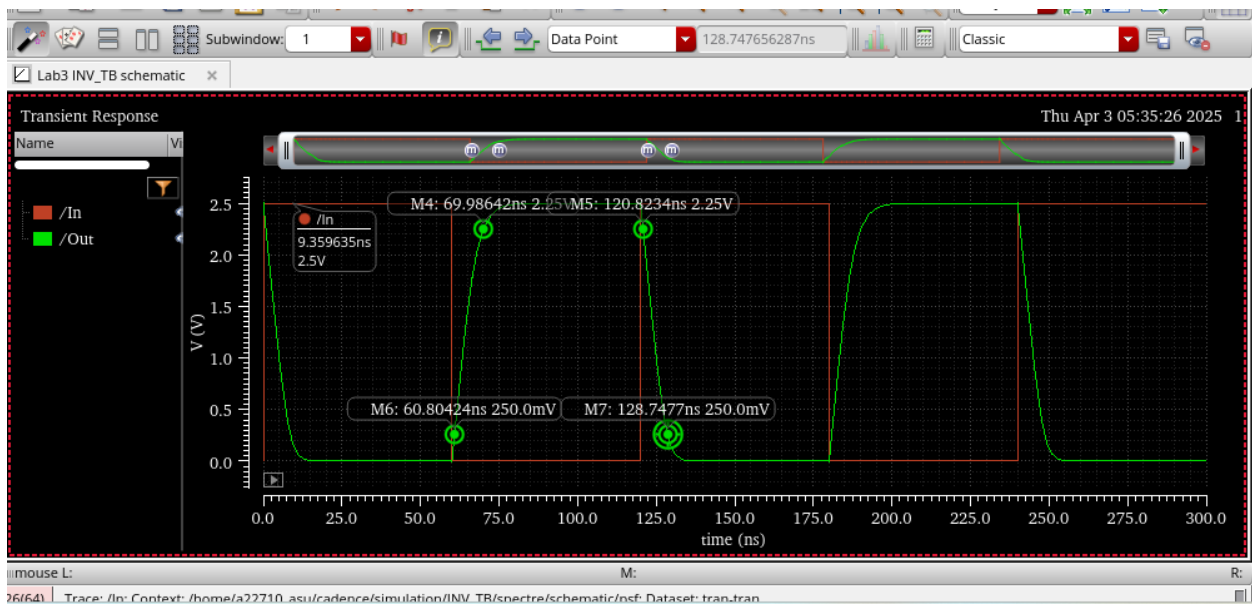


Fig. 4. Inverter Input vs. Output

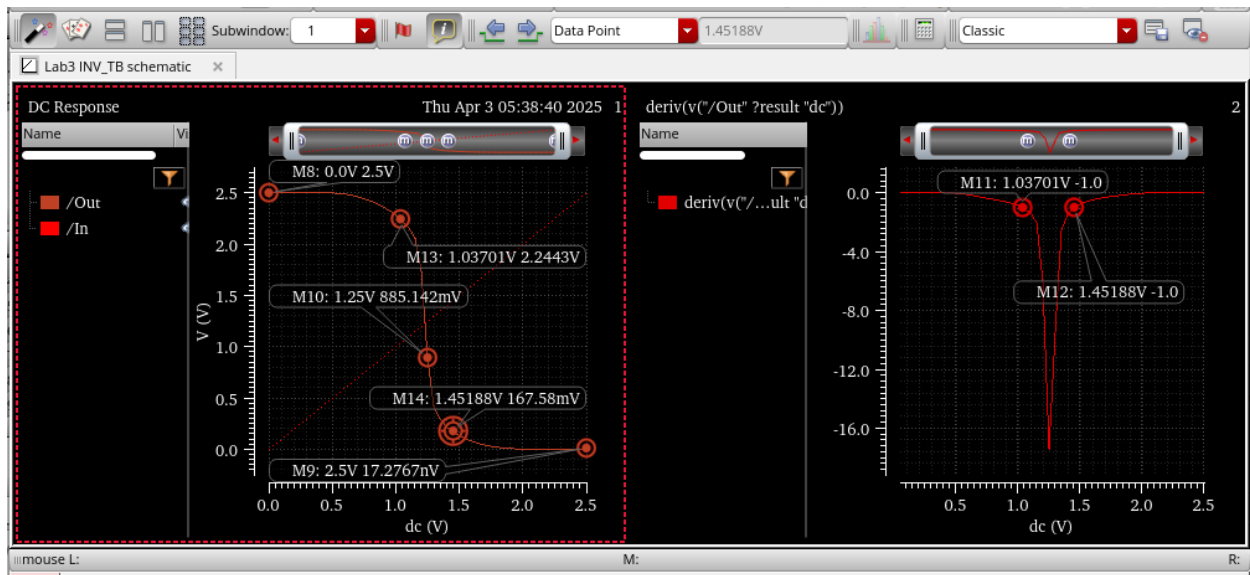


Fig. 5. Inverter Voltage Transfer Characteristic

Table I. Inverter Comparisons Table

Condition	Wp	Tphl	Tplh	Rising	Falling	VM	N _{ML}	N _{MH}
Hand Calculation	1.291u	1.443n	1.754n	N/A	N/A	N/A	N/A	N/A
Tphl=Tplh	1.272u	4.174ns	3.8ns	9.182ns	7.9243s	1.25V	1.037	1.452

Conclusion: There was some discrepancy between where the rise and fall times and the propagation time were equal, and the inverter did not behave exactly as calculated because the actual Wp was slightly wider. The error introduced was probably assuming the inverter had a matched transistor, but in reality the threshold voltages were not equal.

Task 2. NAND Design and Simulation (17.5)

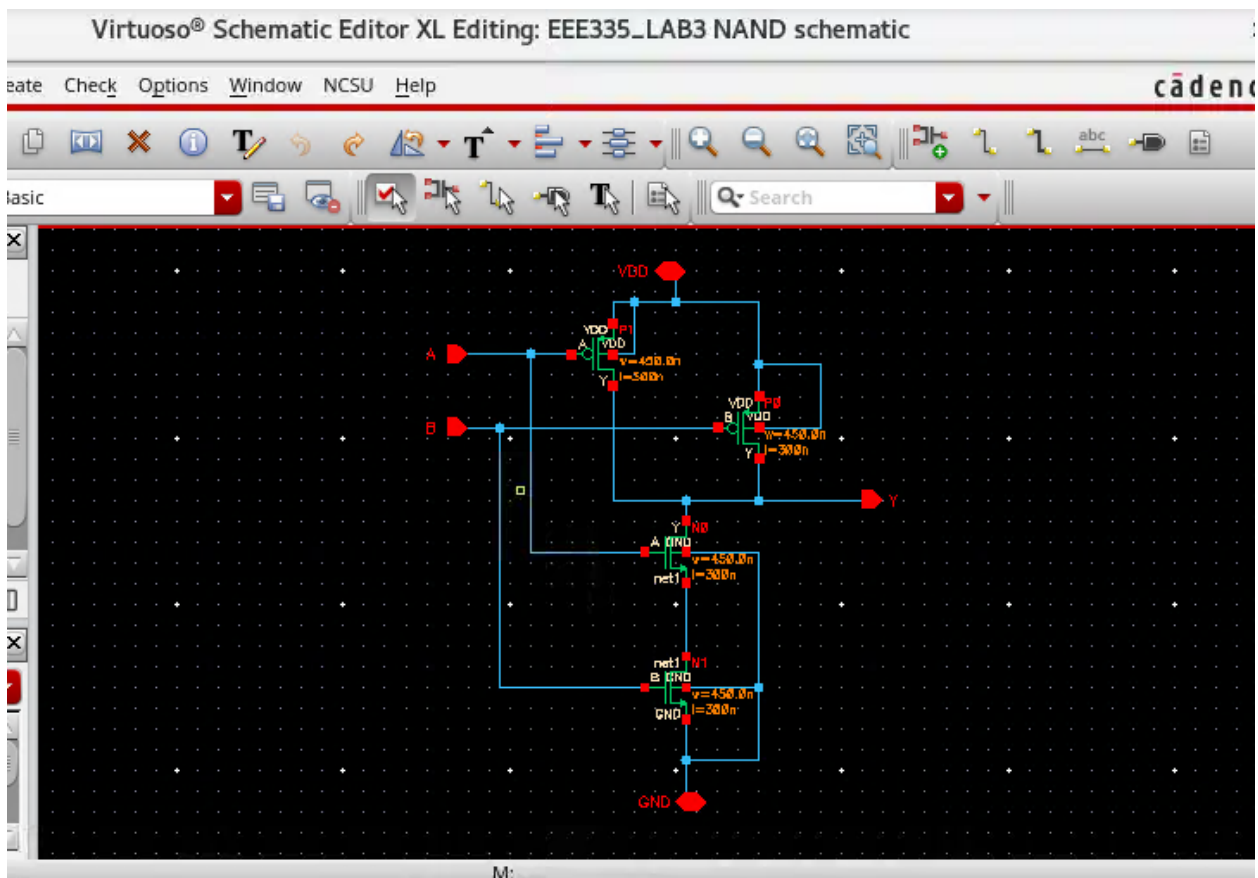


Fig. 6. NAND Schematic Cell

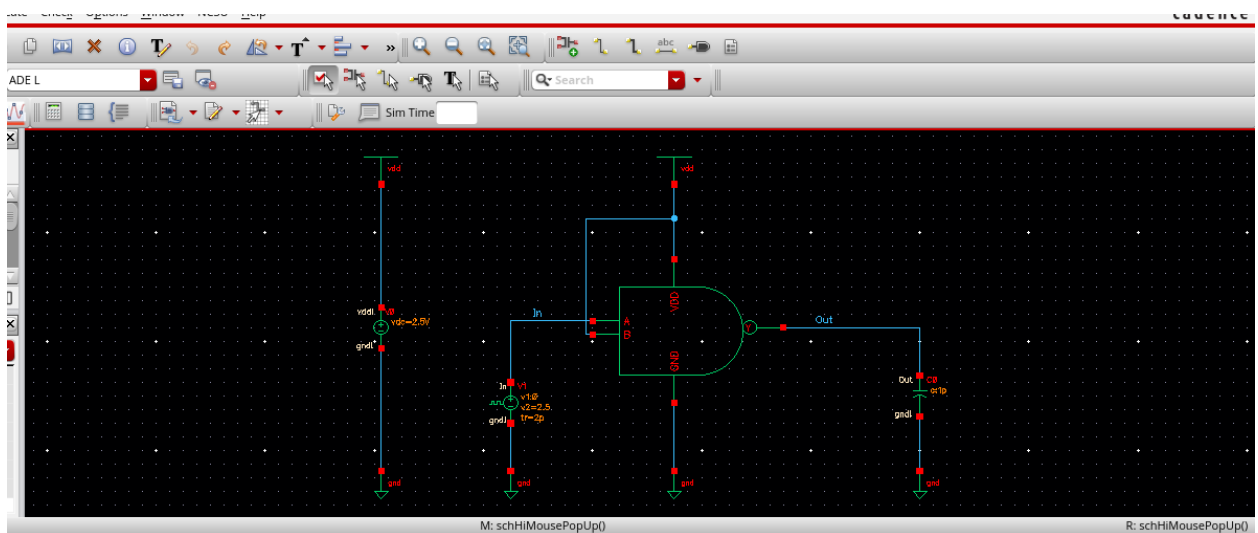


Fig. 7. NAND Test Bench Schematic

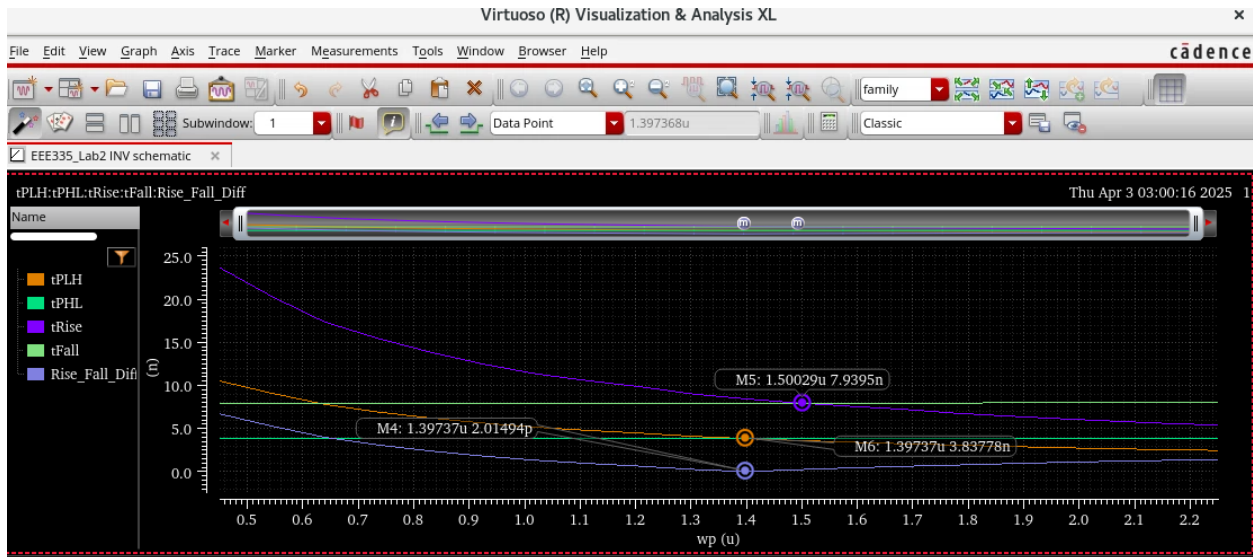
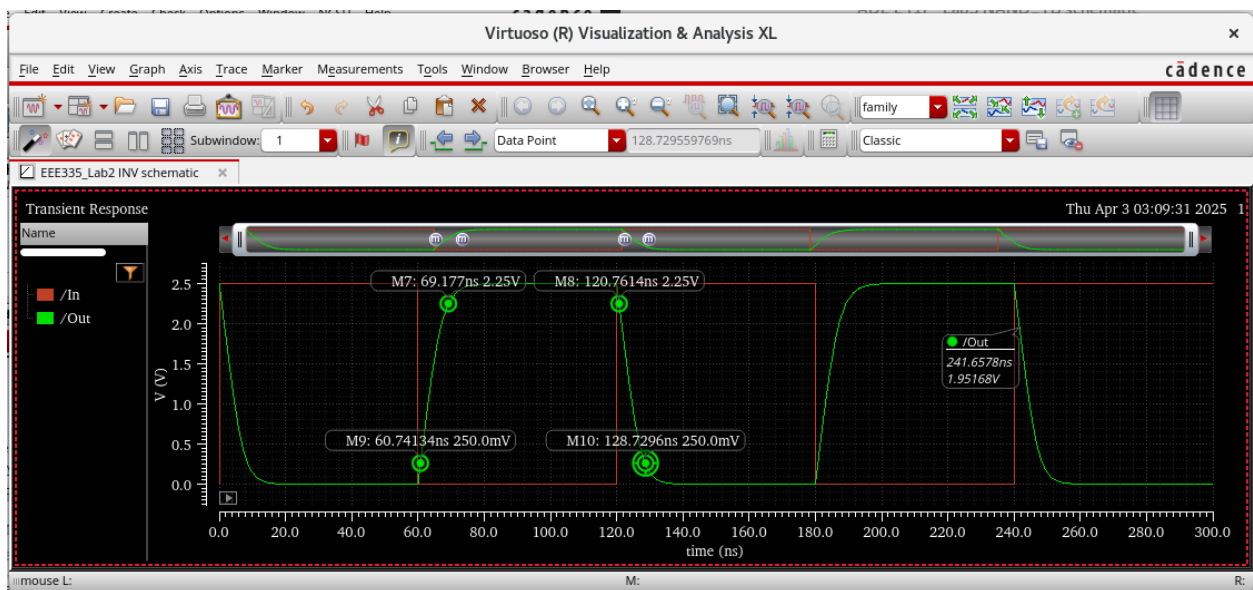


Fig. 8. NAND Parametric Sweep of PMOS Width (t_{pHL} , t_{pLH} , t_r , t_f , Rise Fall Delay)



9a. NAND A vs. Output

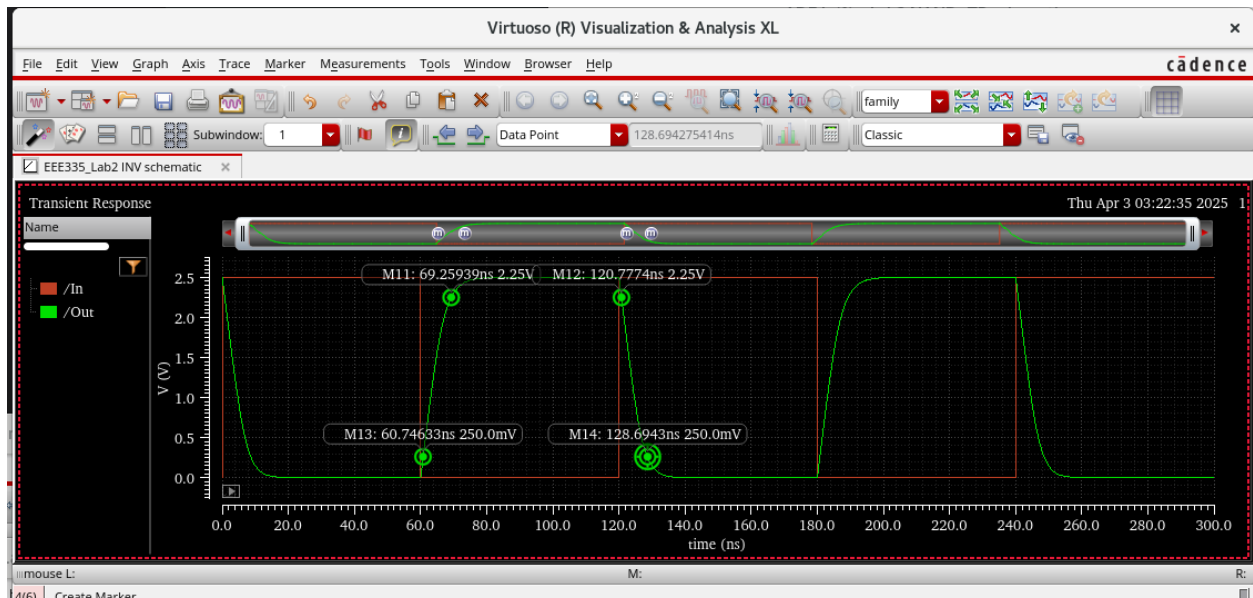


Fig. 9b. NAND B vs. Output

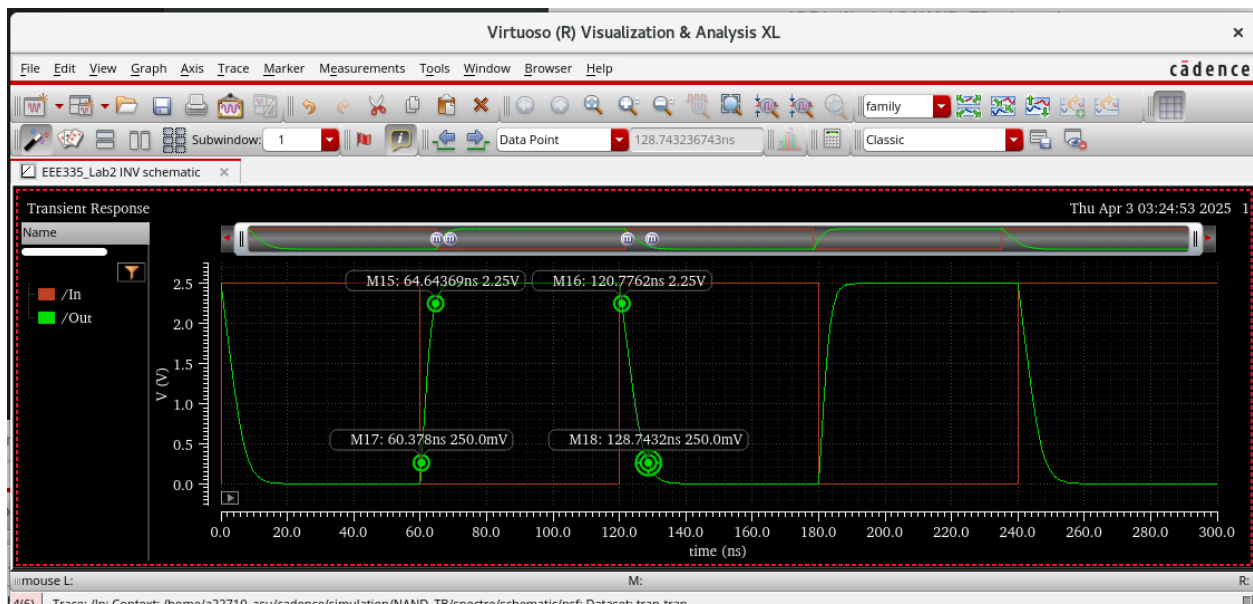


Fig. 9c. NAND A & B vs. Output

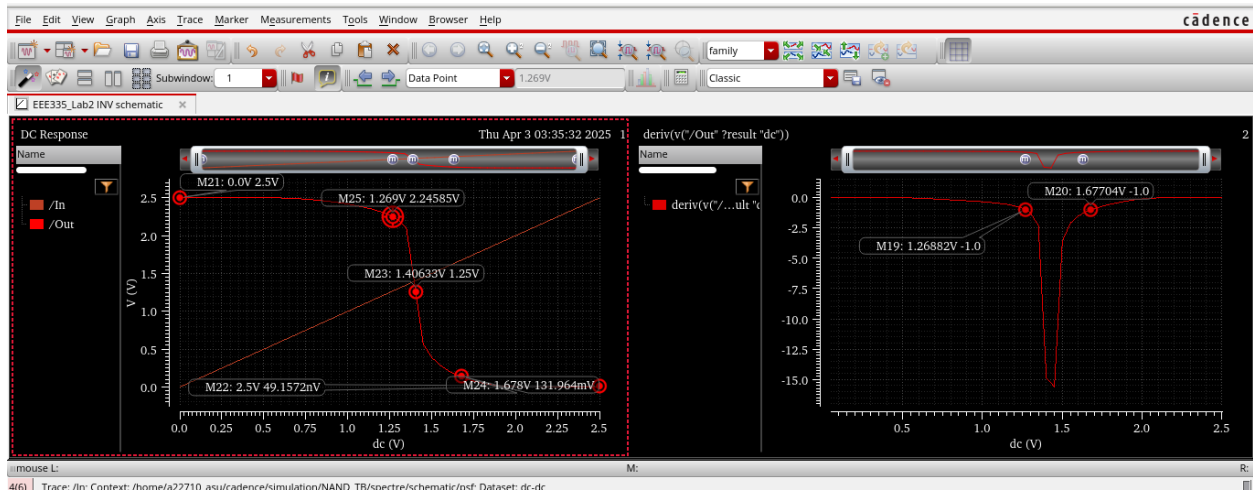


Fig. 10. NAND Voltage Transfer Characteristic

Table II. NAND Comparisons Table

Condition	Wp	Tphl	Tplh	Ris ing	Fall ing	VM	N _{ML}	N _{MH}
Input A	4.5u	3.843n	3.953n	8.436n	7.9682n	1.193	1.146	1.108
Input B	N/A	3.881n	3.608n	8.508n	7.9169n	1.13	986.7m	1.127

Conclusion: I would have expected them to be very similar, but it seems the model takes some input from signal proximity as it would matter in real life readings. Input A is a less balanced inverter because it has the active input, and the difference in magnitudes between the noise margins is nearly double of the active input B. I think this is why input A had a larger NML than input B.

Task 3. NOR Design and Simulation (17.5)

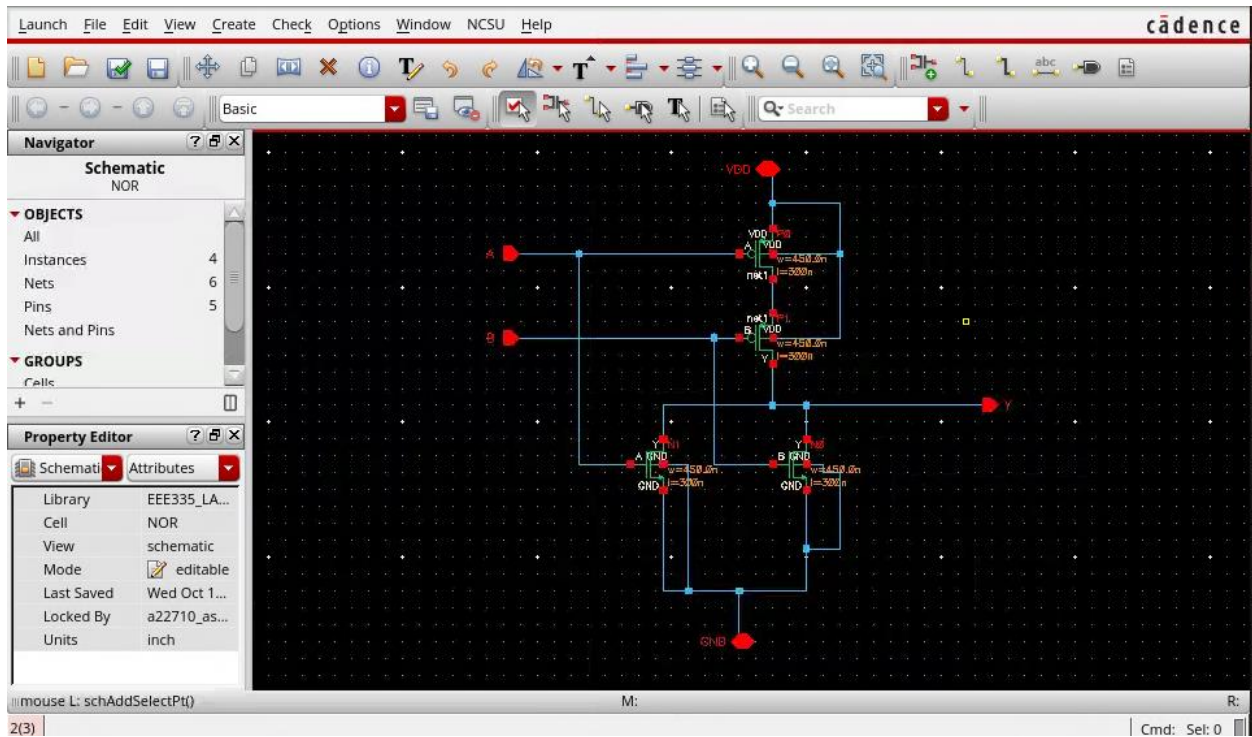


Fig. 11. NOR Schematic Cell

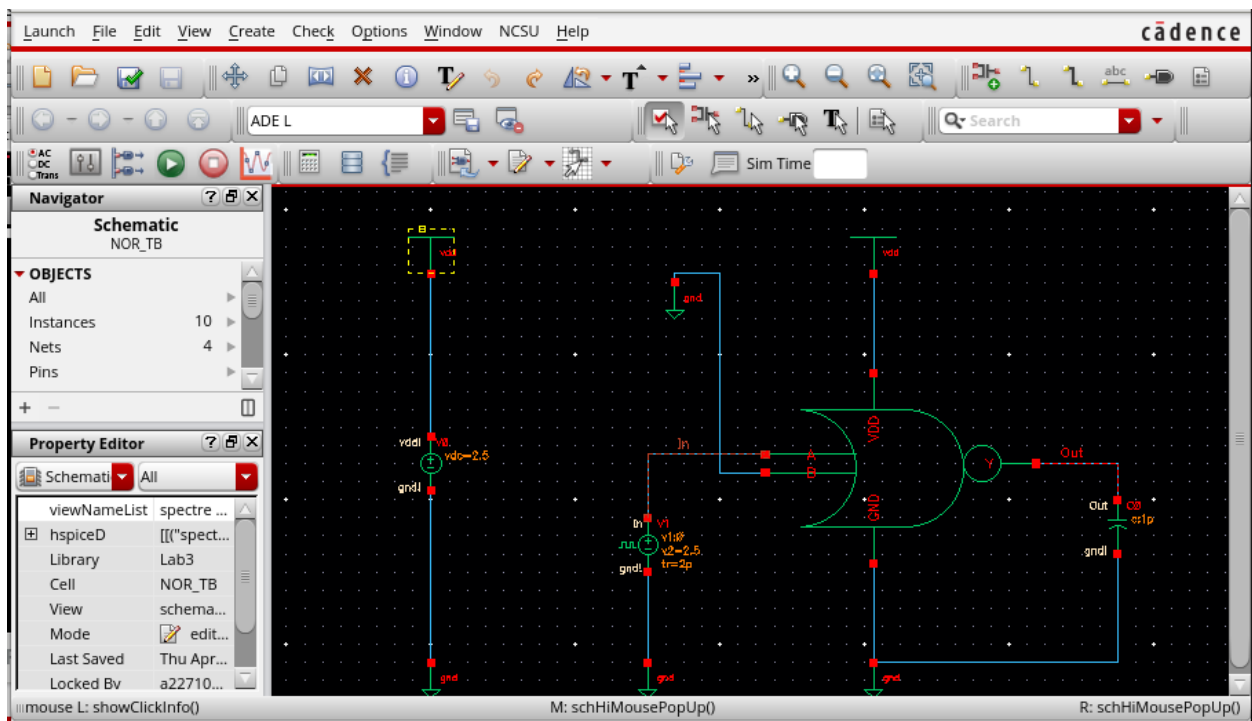


Fig. 12. NOR Test Bench Schematic

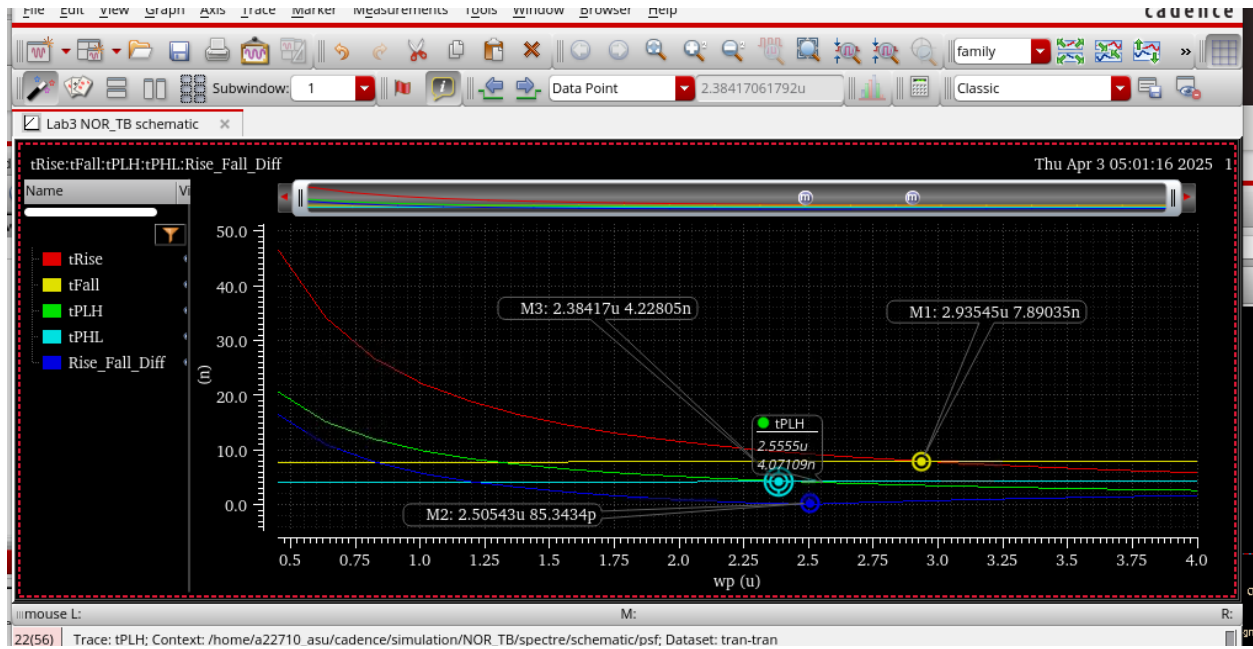


Fig. 13. NOR Parametric Sweep of PMOS Width (t_{pHL} , t_{pLH} , t_r , t_f , Rise Fall Delay)

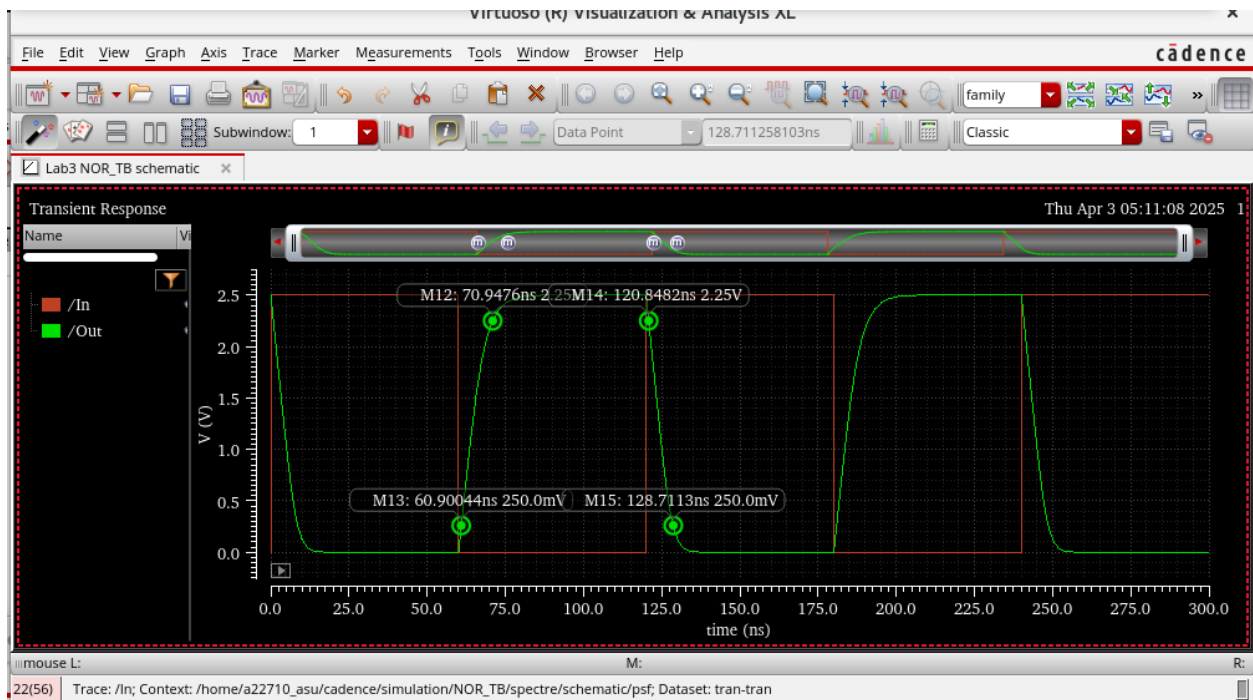


Fig. 14a. NOR A vs. Output

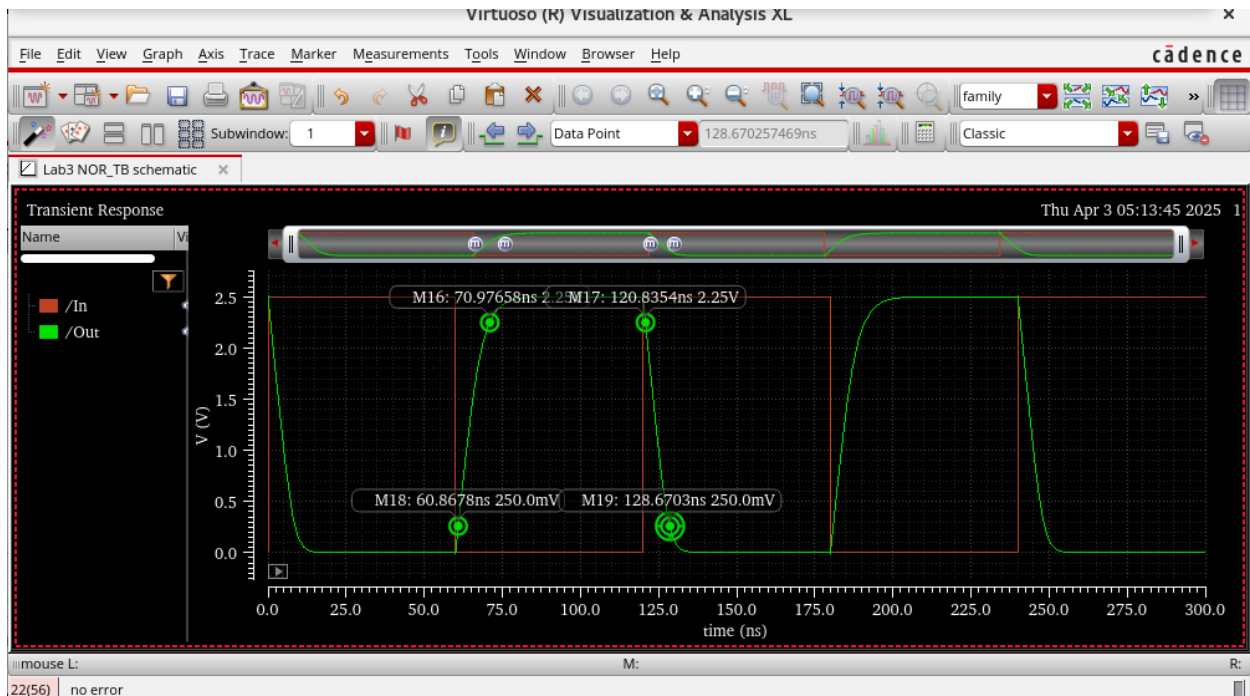
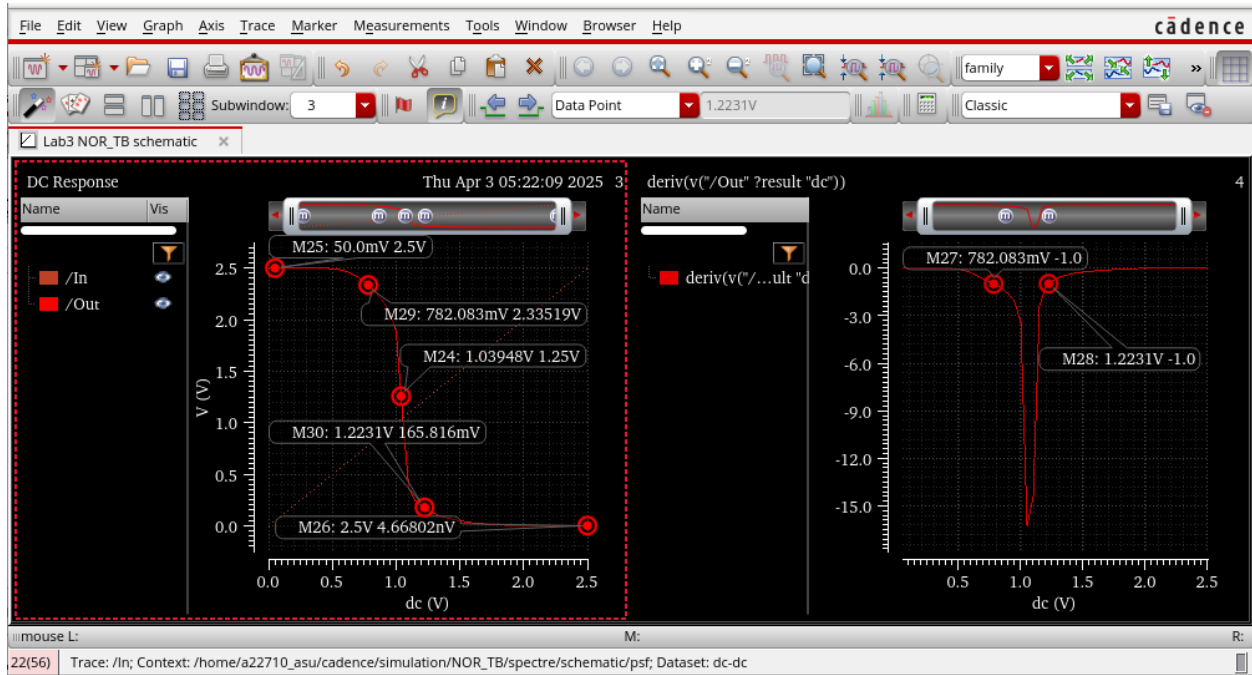


Fig. 14b. NOR B vs. Output



Fig. 14c. NOR A & B vs. Output



1. Fig. 15. NOR Voltage Transfer Characteristic

Table III. NOR Comparisons Table

Condition	Wp	Tphl	Tplh	Rising	Falling	VM	N _{ML}	N _{MH}
Input A	2.25u	3.881n	3.608n	8.436n	7.968n	1.35	1.146	923
Input B	N/A	7.944n	10.231n	8.508n	7.917n	1.13	1.083	1.02

Conclusion: Using inspection techniques, we know the (W/L) for the pmos transistors but be 2p larger than the standard p transistor, meaning the Wp must be twice as large as the minimum Wp needed to have matching transistors. A rule of thumb for Wp vs. Wn is that 2Wp roughly is Wn. This means that for a smallest unit Ntransistor, the Wp is 4*Wn, which was not in the range of the sweep. The Wp parametric sweep did not show a point for the NOR, where the propagation delays and the rise or fall times were equal.